DREADDs and vault delivery: new opportunity screen

ddiscov Round 23 | Evidence cutoff: 6 November 2026 | America/Los_Angeles

Decision

GA002 for refractory focal epilepsy is the strongest new clinical diligence lead in this screen. VaultAAV is the strongest new delivery-platform question. Both earn attention, but neither has a verified, priced investment opportunity that establishes an advantage over cash or the existing conditional development options. This is an information-priority decision, not a finding of negative financial EV and not an authorization to spend.

The user supplied DREADDs and protein vault nanoparticles as seeds. The search therefore followed exact therapeutic programs and delivery claims rather than awarding a high rank to a technology category. The most important changes in belief are: a small human signal makes DREADDs more than a laboratory-tool thesis; a new vault paper gives a concrete antibody-shielding hypothesis; and current ownership plus contrary functional evidence sharply narrow what could be financed by a new participant.

Priority	Opportunity	Present disposition	Main decision-changing uncertainty
1	GA002 / drug-controlled chemogenetic neuromodulation for refractory focal epilepsy	New clinical diligence priority; existing sponsor program	Reproducible clinical benefit, longer safety, and an obtainable economic interest
2	VaultAAV / protein-vault delivery of AAV to address neutralizing antibodies	Platform evidence and rights reserve	Translation beyond cell-level protection and the actual commercial rights chain
3	PSAM4-GlyR sensory-neuron control for pain	Adjacent preclinical reserve	Useful pain relief with acceptable sensory effects and a defined product/rights route
Lower	CCL21-vault for lung cancer	Disease-specific delivery reserve	Carrier-added benefit, contemporary clinical rationale, and a credible product/funding bridge
Hold	PSAM4-GlyR/uPSEM817 for epilepsy	Function-limited research lead	Cellular inhibition did not establish seizure control in the inspected chronic model
Park	NY-ESO-1-vault, bryostatin-vault, antibody-targeted vault-Z	Coverage records, not promoted investments	Immune phenotype or uptake has not established the intended therapeutic outcome

The first three positions compare the usefulness of additional information, conditional on its actual accessibility and cost. They are not calibrated probabilities, full-asset valuations, or a promise that any offered transaction is attractive. An inexpensive, well-defined qualification task can be worthwhile before human efficacy or a signed license exists. Conversely, completing a small technical proof does not finance the obligations that follow it.

1. GA002: a meaningful human signal, with a narrow evidentiary boundary

What changed

The investigator institutions' accounts of the new NEJM report describe three people with unilateral mesial temporal lobe epilepsy and hippocampal sclerosis receiving hM4D(Gi)-based GA002 plus oral clozapine while continuing their background antiseizure therapy. Mean seizure-frequency reduction was **92.6% during postoperative weeks 5-12**. The reported monthly counts were 4.5 to 0.5, 4.0 to zero, and 4.5 to 0.5. Safety was the primary endpoint. Reported adverse events were grade 1 or 2, including one transient, asymptomatic liver-enzyme increase. Follow-up was 12, 8 and 8 months; two breakthrough seizures in one patient occurred during an unplanned interruption of the activating drug. [D01-D03]

This supports a clinically relevant hypothesis: a localized introduced receptor might allow ongoing pharmacological control of the seizure focus without requiring tissue removal. The oral controller is a potential product advantage because the degree of receptor activation can be adjusted. That advantage is about control of activity; stopping a drug does not remove the delivered gene or make all consequences of treatment reversible.

What the signal cannot establish

This was a three-person, uncontrolled pilot. The percentage is neither a randomized treatment effect nor the percentage of patients cured. It cannot establish comparative efficacy against resection, ablation, neurostimulation, or another gene therapy. Low baseline event counts, background medication, seizure variability and effects of the intervention itself all limit causal interpretation. An interruption followed by renewed seizure control is supportive within-person evidence, but an unplanned episode is not a blinded withdrawal comparison.

Absence of a serious event in three people gives little precision about uncommon or delayed harms. Longer observation could also change the apparent durability of benefit. Useful on-treatment control can be valuable; permanent cure is not required for the commercial thesis. It would nevertheless be an error to describe this as a reversible cure or to generalize from a narrow focal-epilepsy population to psychiatric disease, Parkinson's disease, or widespread epilepsy.

Development status and cohort accounting

The November CIBR disclosure reports China registration-stage enrollment and US phase I/II trial permission. Those are institutional development-status reports; the FDA permission letter was not independently inspected and there is no commercial-approval claim. A July 2026 Beijing authority report described six people dosed in the broader investigator-initiated program, including three with temporal-lobe epilepsy and hippocampal sclerosis. The six-person statement and the three-person publication must not be added into nine independent patients or treated as six detailed efficacy records. [D02-D04]

The full NEJM article and supplement were not recovered in this root review. The precise result above is attributed to the investigator-authored institutional account, corroborated at publication and program level. No missing protocol, adverse-event table, patient-level diary or statistical plan was reconstructed from secondary reporting. This access limit remains relevant to a future clinical diligence decision. [D01-D03]

Rights, financing and value capture

GenAns is the existing developer. An August 2026 disclosure from an investor reports a Series A of several hundred million renminbi intended to advance core programs and US/China development. It is not a disclosed valuation, an offer to participate, or proof that every remaining development cost is funded. GA001, mentioned in the same corporate materials, is an optogenetic retinal program and does not add DREADD-treated epilepsy patients. [D04-D05]

For ddiscov, three commercial forms would have different economics: an investment in the existing company, an appropriately scoped license or partnership, or a genuinely differentiated future program. None is assumed available. A public receptor concept or paper alone does not transfer clinical data, manufacturing knowledge, geographic rights, or freedom to operate. The generic availability of an activating medicine does not make the receptor-plus-vector treatment generic or cheap. A published UCL Business patent record, US10525103B2, is an additional reminder that combination-related claims already exist; this screen makes no infringement, available-license or freedom-to-operate finding. [D06]

The plausible upside is a platform that controls abnormal activity in selected neural circuits. The cost is a treatment system with gene-delivery, clinical-procedure, controller-drug and long-term safety obligations. Future value has to be evaluated after those costs and after the royalties, ownership dilution or other sharing required to obtain a claim. Recent funding and expected sponsor-generated results belong in the waiting comparison; they are not a reason to repeat work already being financed elsewhere.

Finite next decision

The useful additional record would connect the exact published clinical population and longer follow-up with the current registered program, a clear commercial counterparty, and a specific participation or contribution. A larger, interpretable clinical result with a credible economic route would raise priority. Failure to reproduce useful seizure control, a material safety problem, or economics that leave too little retained value would lower it. An incomplete record is inconclusive rather than a negative result.

Disposition: first new clinical information priority in this screen. No paid tranche, acquisition, independent reconstruction or clinical activity is authorized.

2. VaultAAV: a concrete delivery hypothesis with an existing ownership history

The exact claim is antibody protection in a cell system

Collins and colleagues' 2026 ACS Synthetic Biology paper reports AAV packaged in recombinant protein vaults and transduction of cultured cells in the presence of neutralizing mouse serum. The paper includes nonpackaged vault-plus-AAV and AAV comparators. Nonpackaged mixtures also improved transduction when neutralizing serum was absent, making increased uptake alone insufficient evidence of encapsulation-specific protection. This is **cell-level proof of principle**, not an in-vivo therapeutic result or human redosing demonstration. [V01-V02]

The attractive hypothesis is an eligibility advantage: a delivery system that works despite otherwise blocking antibodies could make a valuable gene therapy useful for more patients. If it also enabled safe later administration, that could have additional value. Neither clinical property follows automatically from the published cell result. The evidence does not yet establish preserved organ

targeting, useful therapeutic expression, disease rescue, reliable repeat administration, or human immune tolerance.

The fact that vault proteins occur naturally in mammalian cells, including human cells, is a rationale for investigation, not proof that an engineered, externally administered, cargo-bearing product is immunologically invisible. Likewise, evidence for one cargo cannot establish that a vault can deliver every therapeutic protein, RNA, or drug. A carrier has to earn its extra product complexity with a useful advantage over a practical alternative. [V01, C04]

The opportunity is already being commercialized

The paper discloses Cathedral Therapeutics and the investigators' commercial interests. Published application **US20250027112A1** identifies the AAV-vault delivery concept; public metadata show Washington University and a later Regents of the University of California assignment entry. A published application and database status are not a granted worldwide monopoly, a current license, or a freedom-to-operate opinion. [V01, V03]

The inventor's own website reports that Cathedral was acquired in March 2025 by an unnamed biotech and that a new Cathedral commercial entity was created. The current company site presents a proprietary delivery platform. The acquisition is a first-person report: the acquirer, transaction economics, executed licenses and complete chain of title were not independently verified. It would be incorrect to characterize this as an unclaimed academic invention awaiting a new founder. [V04-V05]

The economic question is therefore specific: what commercial field, data access or ownership interest could a new participant obtain, and what would that participant contribute? Corporate existence does not prove complete funding; it does change the burden of showing why the next outside dollar creates retained value.

What would move the decision

An additional public record showing a useful in-vivo or clinical delivery advantage would improve the scientific assessment. The commercially consequential record would establish who controls the relevant rights and whether an attainable participation route exists. Before a larger commitment, the intended therapeutic product, remaining development costs, manufacturing consistency and partner economics would have to be joined into one case.

This does not require a completed human therapy before any small qualification expenditure could be rational. It requires that the proposed expenditure answer a consequential question and connect to a feasible next choice. Repeating the existing cell result in a desktop summary buys no additional information.

Disposition: strongest vault-related platform reserve; second information priority in this seed screen. No disease indication is nominated merely to turn a platform into an apparent drug candidate.

3. Adjacent chemogenetics: PSAM pain merits a separate reserve

PSAM/PSEM systems use engineered ligand-gated channels and are distinct from GPCR-based DREADDs. The human GA002 observations cannot be credited to PSAM programs. Different controllers and receptor configurations also cannot exchange safety or efficacy evidence merely because both are called chemogenetics.

The 2023 Science Translational Medicine work on PSAM4-GlyR reported reduced pain-related behavior in mice and reduced hyperexcitability in human iPSC-derived sensory neurons. Reversible, ligand-dependent pain suppression remained available ten months after delivery in the mouse work. Human neurons in a laboratory are not treated patients. A separate 2023 eNeuro study reported pain-related effects in mice; sensory outcomes differed with the system and targeting context, including reduced innocuous touch in part of the work. These results support a pain hypothesis while preserving a central product tradeoff: meaningful pain relief must be compatible with acceptable ordinary and protective sensation. [P01-P02]

An October 2026 preprint titled Precision cell-state silencing of injured afferents alleviates spontaneous neuropathic pain was located in the original bioRxiv listing. Its full primary text was not obtained in this root review. It is recorded as a new bibliographic lead, not as independently verified therapeutic efficacy or a peer-reviewed replication. [P03]

The commercial appeal is specific peripheral pain control with an adjustable drug component. The unresolved burden includes exact product identity, sustained benefit, delivery practicality, safety, control-drug suitability and usable rights. Research-material access or an academic publication would not establish clinical manufacturing or commercial availability. No company association is assigned on the basis of name similarity, and no current trial or licensable package is assumed.

Disposition: preclinical reserve after GA002 and vaultAAV for this information queue. It could rise if a defined, affordable contribution resolves a useful-function or product uncertainty. No human efficacy is required before that question is considered; no paid qualification scope was priced here.

A separate epilepsy result limits enthusiasm

The Gene Therapy study published online in October 2024, in a 2025 issue, evaluated PSAM4-GlyR with uPSEM817. It found reduced excitability in cells and epileptiform activity in slices, but did **not significantly suppress electrographic seizures** in the inspected chronic mouse epilepsy model. That is a material functional limitation for that exact program, even though the authors discuss possible reasons and future potential. A trend in interictal discharges is not a successful seizure endpoint. [P04]

This does not invalidate every PSAM configuration or the separate GA002 treatment. It does prevent a generic inference that neural silencing in a dish has already translated into robust seizure control. The exact PSAM epilepsy record stays on hold pending a substantively different functional result.

4. CCL21-vault: real animal activity, a weaker human bridge

The 2011 PLOS ONE lung-cancer study reported antitumor activity, with approximately sevenfold lower orthotopic tumor burden and figure-legend groups of ten mice. The relevant in-vivo comparators were empty vaults and diluent. Those arms support activity of the cargo/carrier package, but do not establish therapeutic superiority over free CCL21 or a competing CCL21 delivery product. A related 2020 glioma study used a flank model; it does not demonstrate brain delivery, intracranial disease control, or passage through the blood-brain barrier. [C01-C02]

The most important contrary context is a **different product**: an October 2026 trial of CCL21-modified dendritic cells plus pembrolizumab in advanced lung cancer. It treated 23 people; 19 were efficacy-evaluable. There were **0/19 confirmed objective responses**, and **7/19 had stable disease (36.8%)**. Median progression-free survival was 2.1 months. The response endpoint was not met despite evidence of immune infiltration. This is not a failed vault trial. It weakens the

inference that CCL21-mediated immune recruitment alone establishes meaningful clinical benefit. [C03]

A premanufactured vault product could still have practical advantages over an autologous cell product. Such convenience has commercial value only if the whole product delivers a useful clinical result at an acceptable burden and cost. Lower production complexity cannot substitute for a missing efficacy case, and the negative cell-product result is not proof that no improved vault formulation could work.

UCLA's July 2025 account described production-partner disruption, a revised manufacturing approach, and an approximately $10 million funding need for the proposed CCL21 trial. That was a historical statement, not a verified 2026 financing quote. Earlier company language about safety of an active ingredient is not evidence that people had received vaults. No vault-treated human efficacy outcome was verified in this bounded review. Public cytokine-vault patents also make a clean, available rights package an unresolved question. [C04-C06]

Disposition: lower delivery reserve. A new clinical artifact, a convincingly differentiated product case, or unusually attractive and usable rights could reopen it. Age of the research alone is not a rejection; the controlling issues are the incremental carrier contribution, clinical rationale, remaining cost and retained economics.

5. Additional vault leads screened and parked

NY-ESO-1-vault. A 2020 study showed uptake and maturation of murine dendritic cells. It did not establish patient benefit or tumor control in that publication. This remains an antigen-delivery lead, with no automatic claim that immune-cell maturation produces clinically useful antitumor immunity. The VPI-111 alias was not verified and is not treated as identical. [S01]

Bryostatin-1 in lipophilic vaults. The 2014 study reported HIV-latency activation and immune-activation biomarkers, including work with cells from four human donors and short-term mouse observations. These are not evidence of reservoir elimination or cure. The publication disclosed licensed intellectual property. The result establishes a cargo-delivery observation; it does not resolve the difficult downstream therapeutic endpoint or justify an independent HIV development program here. [S02]

Antibody-targeted vault-Z and a related anti-LADON concept. The 2024 publication demonstrated targeted uptake of fluorescent material into cancer-cell systems, while therapeutic cargo work remained prospective in the discussion. A separate university thesis/abstract lead concerning anti-LADON delivery is not silently converted into completed therapeutic efficacy. These remain technology scouts, not an established siRNA cancer asset. [S03-S04]

The lung and flank-glioma CCL21 records are one cargo-platform family with distinct model claims. Different activating ligands are not automatically separate DREADD drug opportunities. The lower-priority scouts demonstrate search coverage; they should not inflate the number of comparably investable programs.

6. Adjacent commercial and clinical watchlist

The following are lower-confidence discovery leads, separate from the eight screened records. They do not increase the count of validated programs or globally novel families.

Rutgers retinal DREADD licensing family. The university advertises a retinal hM3Dq approach for depression/circadian disorders under S2016-155. A related listing, 2026-025, repeats that

rationale. A March 2026 listing, MPS2016-155, extends the approach to Alzheimer's-related learning and memory findings in rats. All advertise licensing or research collaboration. The displayed animal behavior does not establish human depression or Alzheimer's benefit, and the three listings are not treated as three independent assets. Their relationship, exact patents, current license availability and supporting primary study package remain to be reconciled. This is the clearest explicit university licensing lead found in the adjacent search, but its clinical evidence is weaker than GA002's. [W01-W03]

GA008 pain program. The July authority disclosure names GA008 for refractory neuropathic pain and describes prospective China/US filings. It establishes a sponsor pipeline lead, not completed filings, clinical efficacy or accessible rights. It remains distinct from the academic PSAM work. [D04]

Unresolved registry identities. NCT06596681 surfaced as a refractory-pain "GA" lead; NCT07085195 and NCT07533591 surfaced as STP001 and CGTPD Parkinson's leads. The original registry pages did not expose their structured records during this pass. These discovery labels are therefore provisional: no current recruitment status, cohort size or outcome is adopted. In particular, the unmapped "GA" label is not silently equated with GA008 or GA002. Primary identity confirmation is the next step before substantive screening. [W04-W06]

7. Comparison with the existing portfolio and cash

The retained Round 22 conditional development sequence is APR-246, RM4404, BLU-3797, NDC-9009, mIGF-Gel, exact 8015-P2 and AAT. It is provenance for previously defined information scopes, not a measured financial ordering over every new technology. The earlier 55/15/10 effort allocation does not control this screen, and completed public-source passes are not reopened merely because new seeds appeared.

GA002 supplies a more direct human signal than a cell-only delivery experiment. That supports its information priority. It does not prove that investing at an unknown valuation in an existing gene-therapy company beats a smaller, rights-defined project. Conversely, vaultAAV could have broad platform upside even while its evidence is earlier. Breadth of possible applications does not itself increase the fraction of future value available to this contributor.

The next marginal dollar should be attached to a concrete choice: what information or right is purchased, what decision a positive or negative answer changes, how much additional capital follows a positive result, and how much of the eventual upside is retained. Compare those choices with waiting for existing sponsors' results and with doing nothing. No success probabilities, hypothetical acquisition prices, commercial forecasts, cash yields or simulated returns were inserted into this screen.

Accordingly, **no new development commitment is promoted today**. That is an unresolved priced comparison, not a demonstration that all these programs have negative EV. A later low-cost option or informative qualification scope could outrank cash. A much larger company commitment would require a different case even if it began with the same paper.

8. Novelty, provenance and stopping boundary

The baseline was GitHub main at e65d88841393132b398dcfe6260b723068a179a9, with tree 0490665ccbe8780e3edb4e335ca381a947a56031. The full 4,729,680-byte cumulative register was recovered and independently matched to its published Git blob. Its 622 original coverage rows were retained. The register, screen index, shortlist and discovery crosswalks were checked alongside complete extracted text of the twelve recovered fresh-screen PDFs B-M.

No direct queried DREADD, PSAM, vault, GA002, CCL21-vault or associated principal alias was found in that inspected corpus. **Screens N-Q and any later completed screens were not recovered.** The novelty claim is therefore limited to the inspected repository and recovered B-M files. The new records remain provisional against the unrecovered screens. An existing AAV indication with a new carrier is an expansion of an earlier program, not automatically a new disease opportunity.

The present public-source pass resolves the leading claim boundaries and identifies the next consequential records. It stops before inaccessible private records, unverified commercial terms or new scientific work. Nothing here requests such records from others or assumes they exist. All restrictions remain: no outreach, purchases, recruitment, experiments, simulations, material receipt or applications. This round contains high-level investment analysis and evidence accounting only; there are no experimental procedures or operational biological instructions.

The accompanying source register records exact URLs, publication dates where established, source type, access limits and claim scope. Company or institutional statements remain attributed, patent records do not become legal opinions, and multiple accounts of the same underlying participants or experiment do not become independent replications.

Sources and access limits

Source IDs in the report link to the entries below. Original papers, institutional disclosures, company statements and patents retain their distinct evidentiary status. Several records are locations or bibliographic leads whose full contents could not be recovered.

D01 | Dai et al., Chemogenetic Neuromodulation for Refractory Epilepsy

2026-11-05 issue | Original clinical report

Publication identified; direct article and supplement not retrieved by root. Quantitative claims attributed to D02, not represented as a full-article audit. Open original source

D02 | Capital Medical University / Xuanwu Hospital investigator account

2026-11-05 | Investigator-authored institutional disclosure

Full page read. Three patients; weeks 5-12; 92.6%; background treatment; safety and 12/8/8-month follow-up. Same cohort as D01. Open original source

D03 | CIBR first-human chemogenetic report and program status

2026-11-05 | Investigator institution

Indexed primary text plus English publication notice. China enrollment and US trial permission remain attributed; regulator letter not inspected. Open original source

D04 | Beijing science authority: GA002 trial launch

2026-07-08 | Official authority reproducing CIBR program disclosure

Full page read. Six IIT recipients include three hippocampal-sclerosis patients; not six detailed outcomes or a cohort additive to D01. GA001 optogenetic; GA008 filings were prospective. Open original source

D05 | ZGC investor: GenAns Series A

2026-08-05 | Investor primary disclosure

Indexed original release read. Several hundred million RMB; no precise amount, valuation, accessible offer or fully funded development assertion. Open original source

D06 | US10525103B2: combined modified-receptor vector and agonist for seizures

2020-01-07 publication | Published patent

Public document and UCL Business assignee metadata inspected. No claim construction, infringement, license availability or freedom-to-operate conclusion. Open original source

P01 | Perez-Sanchez et al., A humanized chemogenetic system inhibits murine pain-related behavior and hyperactivity in human sensory neurons

2023-10-04 | Original preclinical paper

Indexed primary PubMed/PMC article content reviewed; direct PMC access challenged in root. Mouse pain and human-cell work, not treated people. PMID 37792955; PMC7615191. Open original source

P02 | Haroun et al., Chemogenetic Silencing of NaV1.8-Positive Sensory Neurons Reverses Chronic Neuropathic and Bone Cancer Pain

2023 | Original preclinical paper

Primary abstract/article excerpts reviewed. Keep AAV and transgenic contexts and sensory outcomes distinct; PMID 37679042; PMC10523839. Open original source

P03 | Cooper et al., Precision cell-state silencing of injured afferents alleviates spontaneous neuropathic pain

2026-10-15 v1 | Preprint bibliographic lead

Title/DOI/date identified in original bioRxiv listing. Full primary preprint not recovered; no quantitative efficacy or specificity claim adopted. Not peer reviewed. Open original source

P04 | Gonzalez-Ramos et al., Chemogenetics with PSAM4-GlyR decreases excitability and epileptiform activity in epileptic hippocampus

2024-10-25 online; 2025 issue | Original preclinical paper

Indexed publisher/PMC body and concluding statements reviewed. In-vivo electrographic seizure suppression not significant despite cell/slice effects. PMC11946892. Open original source

V01 | Collins et al., Encapsulation of AAVs into Protein Vault Nanoparticles as a Novel Solution to Gene Therapy’s Neutralizing Antibody Problem

2026-08-13 online; 2026-10-16 issue | Original preclinical paper

Indexed ACS body, comparators and conflict disclosure read; direct publisher opening blocked. Cell transduction in neutralizing mouse serum, not in-vivo therapeutic or human repeat-dose success. Open original source

V02 | PubMed record for the vaultAAV paper

2026-10-16 issue | Bibliographic record with author abstract

Metadata and abstract read. Same underlying study as V01, not independent replication. Open original source

V03 | US20250027112A1: Gene therapy delivery compositions and methods of use thereof

2025-01-23 publication; 2023-07-17 priority | Published patent application

Published document and indexed assignment metadata read. Washington University and later Regents of UC entry; no executed license/current clean-title/FTO opinion. Avoid unverified normalized campus label. Open original source

V04 | Logan Thrasher Collins: Cathedral ownership history

Undated page, accessed 2026-11-06 | Inventor first-person account

Full page read. Reports March 2025 acquisition by unnamed biotech and reconstituted entity. No independent acquirer, terms or title verification. Open original source

V05 | Cathedral Therapeutics current website

Undated page, accessed 2026-11-06 | Company disclosure

Current platform and team page read. Marketing claims are not clinical data or validated repeat dosing. Open original source

C01 | Kar et al., Novel CCL21-Vault Nanocapsule Intratumoral Delivery Inhibits Lung Cancer Growth

2011-05-03 | Original mouse study

Full body and figure legends inspected. Ten mice/group in relevant legends; empty-vault/diluent comparison. No free-cargo in-vivo efficacy arm in those contrasts. Open original source

C02 | Voth et al., CCL21-coupled vaults in a glioma model

2020-04-09 online | Original study author abstract

Primary abstract reviewed. Flank model, not intracranial disease/BBB demonstration. Ambiguous free-cargo arm punctuation not silently resolved. DOI 10.1007/s11060-020-03479-8. Open original source

C03 | Oh et al., CCL21-modified dendritic cells with pembrolizumab in NSCLC

2026-10-04 | Original phase I clinical report

Full clinical results read. 23 treated, 19 efficacy-evaluable, 0/19 responses, 7/19 stable disease; different product from vaults. NCT03546361. Open original source

C04 | UCLA Health: How a scientist’s lifelong obsession with vaults could transform medicine

2025-07-10 | Institutional feature with investigator/company accounts

Full page read. Historical manufacturing interruption and approximately $10m prospective trial financing need; not a 2026 quote or verified clinical start. Open original source

C05 | Vault Pharma appoints Oliver Foellmer: historical VPI-101 status

2017 | Company original press release

Reviewed in independent vault source pass. Active-ingredient phase I wording and future trial plans do not show vault-treated patients. Open original source

C06 | US11793860B2: cytokine-vault patent family

2023 grant publication | Published patent

Public family/assignee record inspected in independent pass. No legal claim construction, available license or FTO established. Open original source

S01 | Nagasawa et al., NY-ESO-1 vault nanoparticles induce dendritic-cell maturation

2020-05-07 online | Original study author abstract

Primary abstract reviewed; murine dendritic-cell maturation, not clinical or tumor efficacy. DOI 10.1007/s11060-020-03472-1. VPI-111 alias not confirmed. Open original source

S02 | Buehler et al., Bioengineered Vaults: Self-Assembling Protein Shell-Lipophilic Core Nanoparticles

2014-07-25 online | Original preclinical paper

Full text reviewed. Latency/activation biomarkers; human donor cells are not treated patients; no reservoir elimination. DOI 10.1021/nn5002694. Open original source

S03 | Antibody-targeted vault-Z delivery paper

2024 | Original preclinical paper

Independent pass inspected primary article; uptake/fluorescent cargo result. Therapeutic cargo work prospective; PMC11203631. Open original source

S04 | University abstract concerning anti-LADON vault delivery

2023 abstract | University repository abstract

Independent pass found a proposed siRNA delivery direction; not treated as verified therapeutic efficacy or an alias of S03. Open original source

W01 | Rutgers S2016-155: retinal DREADD neuropsychiatric approach

2018-05-07 listing | University licensing disclosure

Indexed original listing read; direct page returned 403. Rat light-deprivation behavior and licensing advertised; current availability and exact patent scope not confirmed. Open original source

W02 | Rutgers 2026-025: Chemogenetic Treatment for Neurodegenerative and Neuropsychiatric Disorders

2025-09-10 displayed on listing | University licensing disclosure

Indexed original listing read. Similar retinal hM3Dq rationale to W01; not counted as an independent asset or replication. Index/listing dates differ. Open original source

W03 | Rutgers MPS2016-155: intra-ocular neuropsychiatric method / Alzheimer’s-related rat behavior

2026-03-04 listing | University licensing disclosure

Indexed original listing read. Rat learning/memory claims and licensing advertised. No treated-human evidence or independent primary-study audit; relationship to W01/W02 and current rights unconfirmed. Open original source

W04 | NCT06596681: unresolved GA pain registry lead

Current record not extractable | Primary registry location; unverified structured record

Registry page exposed only interface text. GA pain identity is a provisional discovery label, not verified mapping to GA008/GA002; no recruitment, enrollment or outcome claim adopted. Open original source

W05 | NCT07085195: provisional STP001 Parkinson’s registry lead

Current record not extractable | Primary registry location; unverified structured record

Registry page exposed only interface text. Discovery label requires primary confirmation; no recruitment, enrollment, sponsor-rights or outcome claim adopted. Open original source

W06 | NCT07533591: provisional CGTPD Parkinson’s registry lead

Current record not extractable | Primary registry location; unverified structured record

Registry page exposed only interface text. Discovery label requires primary confirmation; no recruitment, enrollment, sponsor-rights or outcome claim adopted. Open original source